SNN implementation on MNIST Dataset

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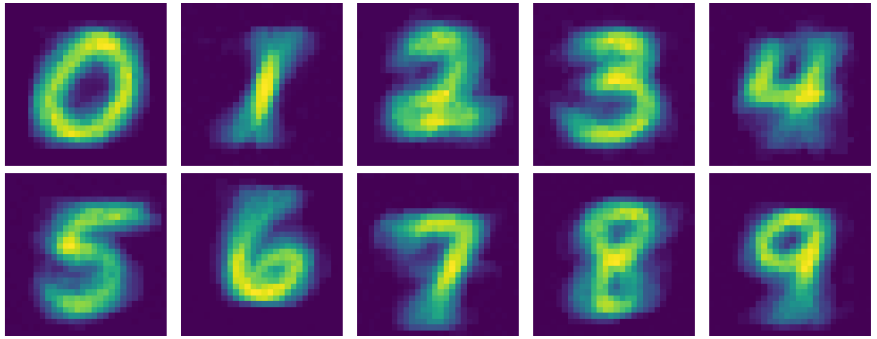
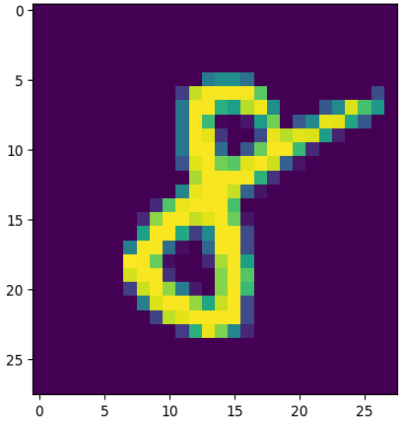
Overview

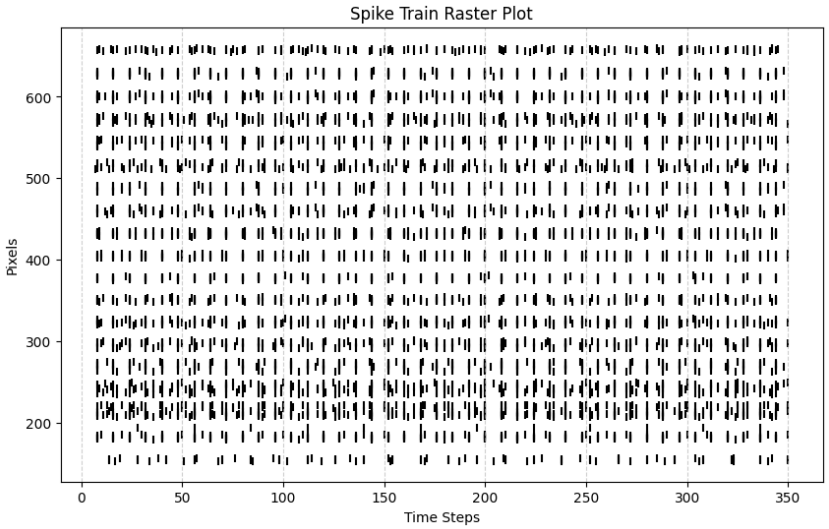
- Achieved approx. 78 – 80 % accuracy on random train sets (2000 images).
- Trained on 400 images. With 320 output neurons.
- Takes roughly 40 minutes to train.
- Working on improving accuracy with varying the parameters, and testing their affect.
- Working on making the weights non linear.



Learned synapses or weights

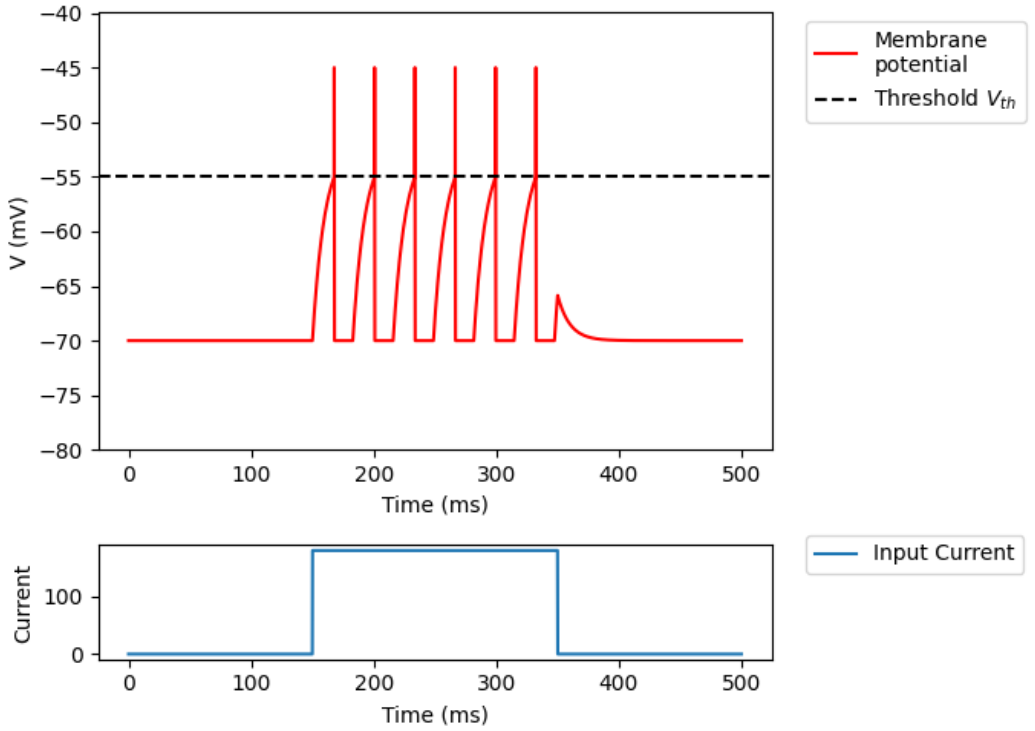


Example Input



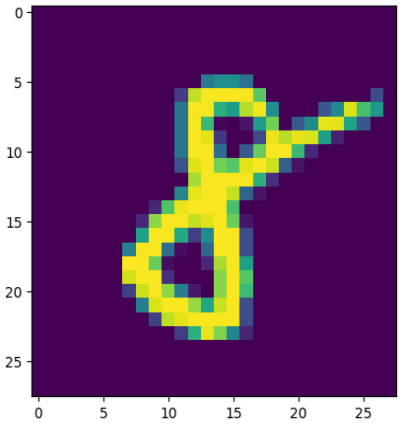
Neuron Model: Leaky Integrate and Fire

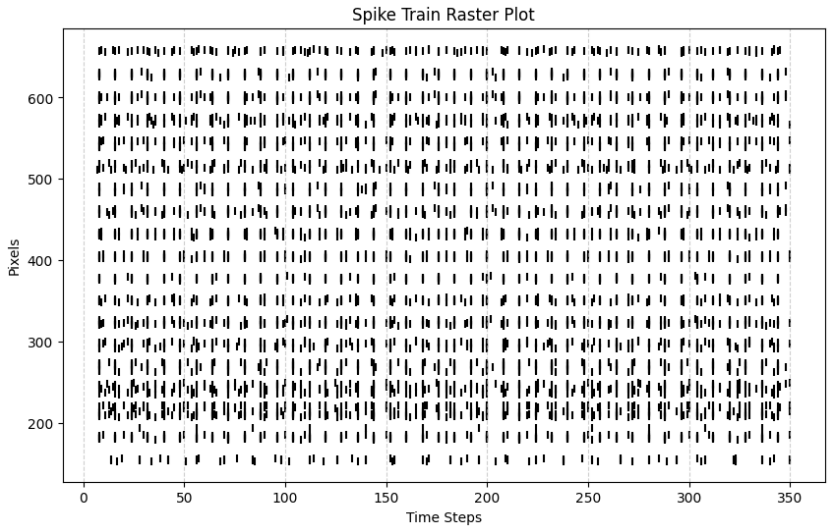
- Simulated a pulse current through a LIF Neuron.
- The neuron slowly builds up the voltage, when it reaches threshold voltage it fires.



Data loading and encoding into spike trains

- Equal number of training images for each class has been used.
- The images are frequency encoded. There is a set training duration for each image (after being normalized), and the intensity value of each pixel (0 – 255) affects the firing of that pixel (neuron) linearly.
- Bright pixels have higher frequency and darker pixels have lower frequencies.
- Then the spiking interval for each pixel is calculated and finally a spike train is made.





Training Process

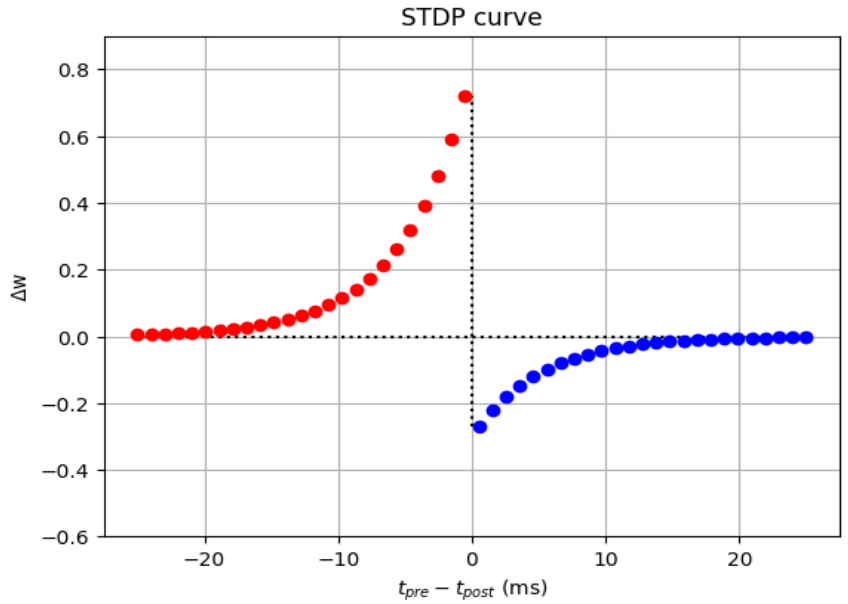
- for each training image, correct label:
  - get the spike train of the image
  - for each timestamp of training duration (350 frames)
    - pass the spike train through the network.
      - check which neurons fired and contributing neurons, and strengthen them (using stdp)
      - weaken the other neurons (using stdp)
    - track a count of the number of times each neuron fires.
  - set the highest firing neuron with the correct label. [neuron lookup table]

The general STDP learning rule for the synapses is given by:

$$\Delta w = \begin{cases} A^{+} e^{\frac{\Delta t}{\tau^{+}}} & \text{if } \Delta t < 0 \\ A^{-} e^{\frac{-\Delta t}{\tau^{-}}} & \text{if } \Delta t > 0 \end{cases} \tag{6}$$

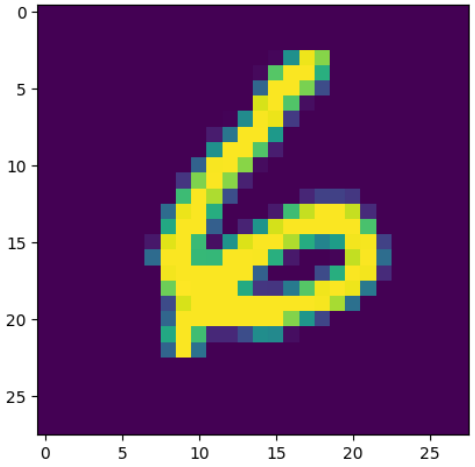
Where, $\Delta t = t_{pre} - t_{post}$. Here the goal is that weights are changed such that if the pre-synaptic spike occurs before the post-synaptic spike, the synapse is strengthened else it is weakened. This ensures that the next (after this training iteration) post-synaptic spike occurs closer to the pre-synaptic spike. The weights are then changed according to the relaxation rule:

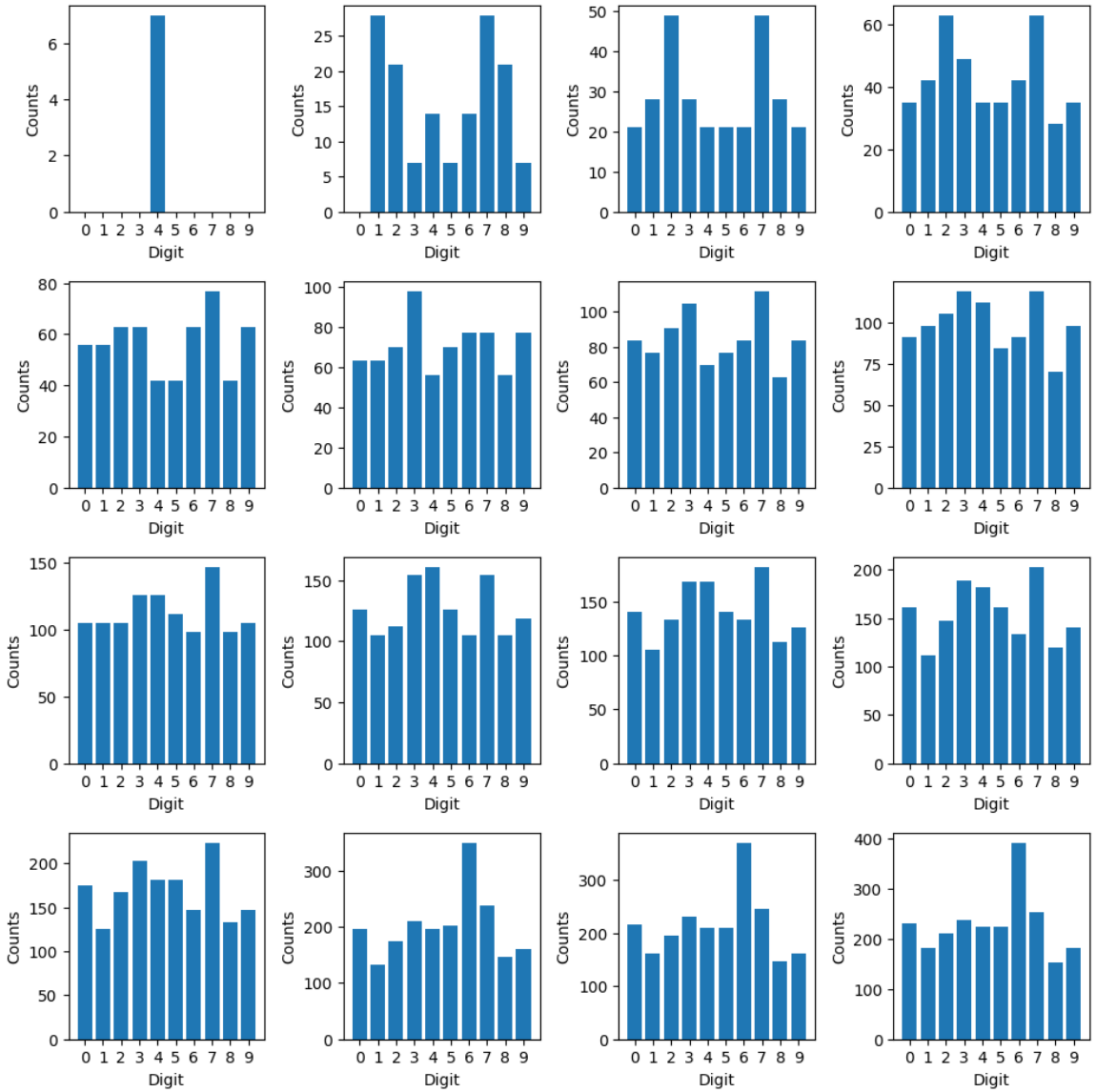
$$w_{new} = \begin{cases} w_{old} + \eta \Delta w (w_{max} - w_{old}) & \text{if } \Delta w \geq 0 \\ w_{old} + \eta \Delta w (w_{old} - w_{min}) & \text{if } \Delta w < 0 \end{cases} \tag{7}$$



Testing Process

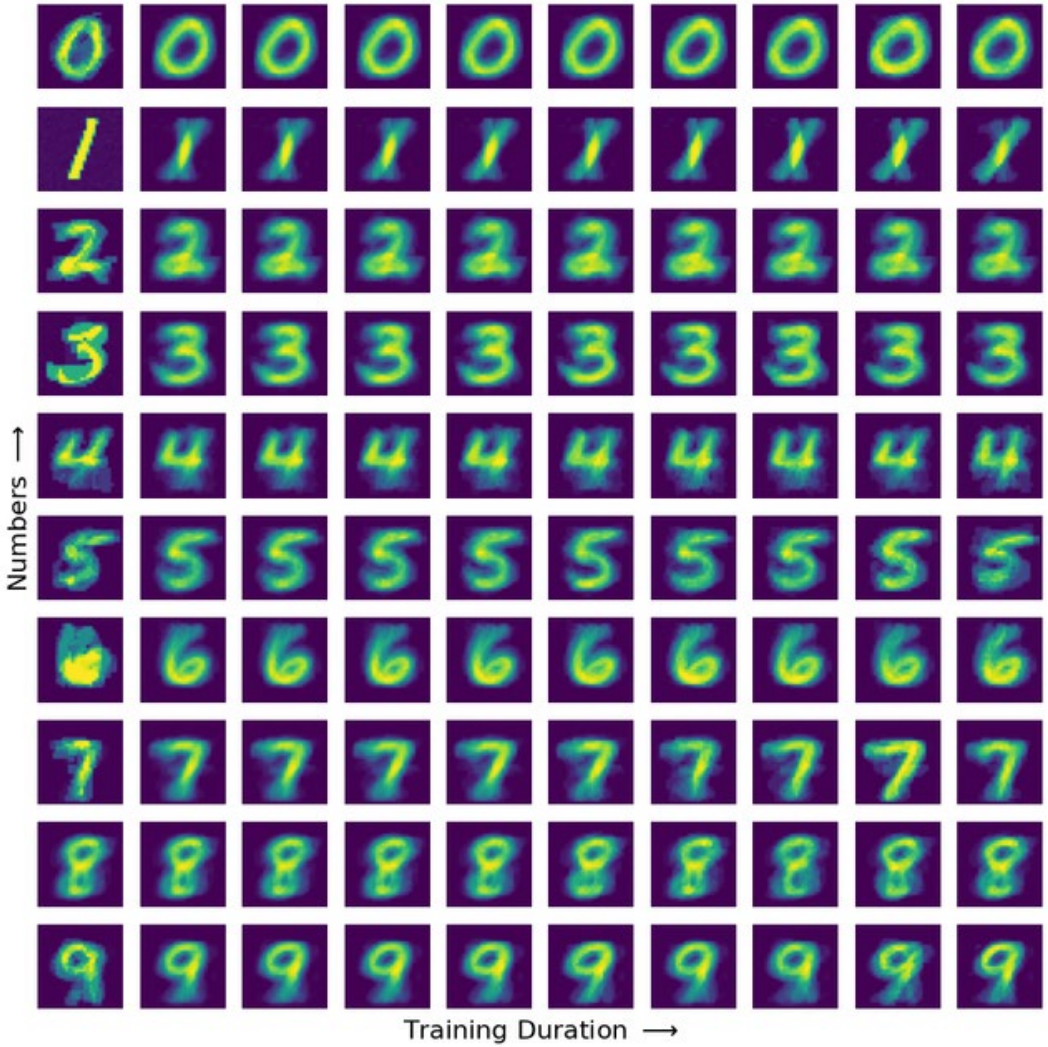
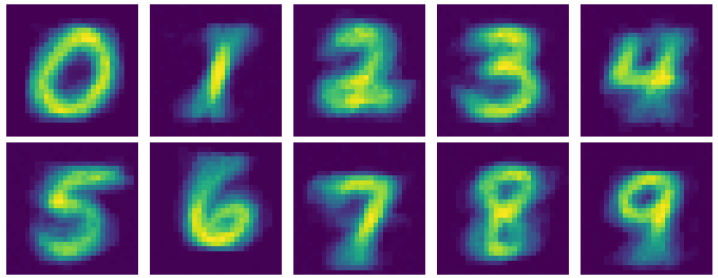
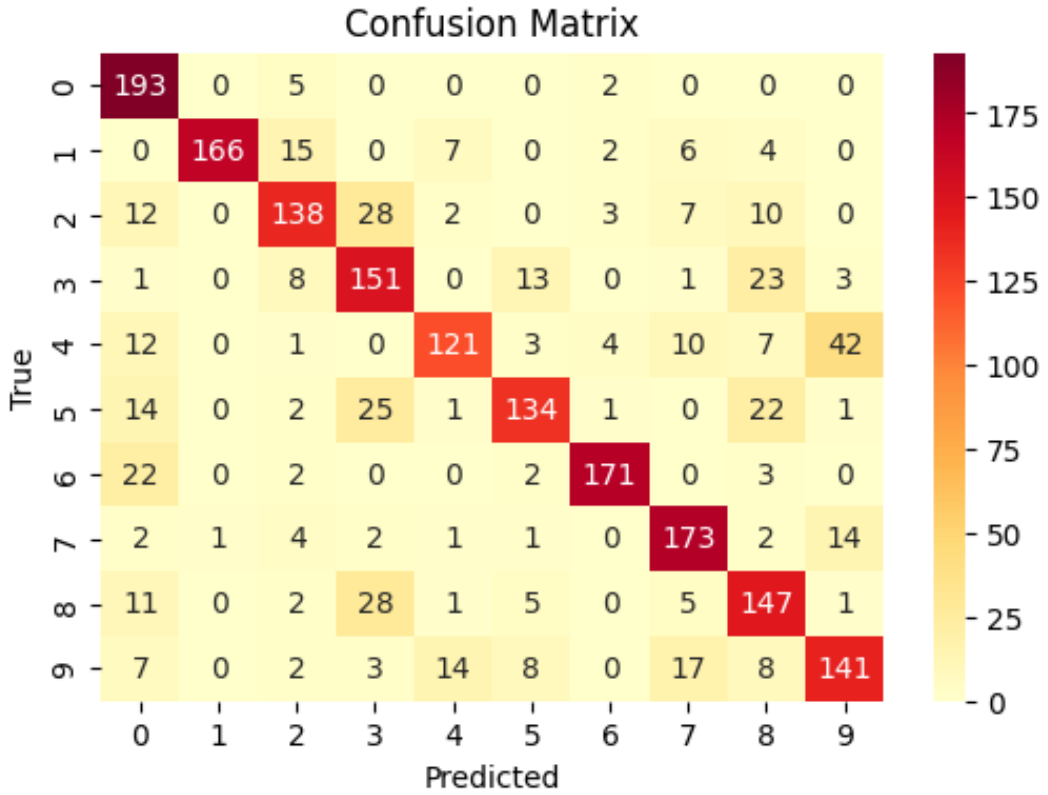
- for each test image:
  - get spike train
  - pass it through the network, and get the neuron that fires the most
  - report the label from the previously determined [neuron lookup table]





Results (Confusion Matrix)

- Accuracy of 78 – 80 % on entirely new images.



Final synapses